# Rapid Prototyping

## *Additive Manufacturing*

# Rapid Prototyping for Polymers

***Rapid prototyping*** involves using additive manufacturing to make a complete part or component in a single operation.

There are several different rapid prototyping processes used to make parts from polymers, including ***fused deposition modelling (FDM)*** and ***stereolithography (SLA)***.

# Rapid Prototyping for Polymers

***Advantages of rapid prototyping:***

A major advantage of rapid prototyping is that, compared to conventional workshop processes, a *product is made in a single operation from a 3D computer aided design model*.

For complex parts, this can be carried out more quickly than the several conventional machining operations needed to make the part.

This allows for ***rapid evaluation*** of whether, for example, a product may fit in the available space *(e.g. a turbocharger in an engine compartment).*

# Rapid Prototyping for Polymers

***Disadvantages of rapid prototyping:***

A disadvantage of rapid prototyping is the ***limited range of materials*** that can be used.

The most common processes can be used only with certain types of ***polymer***; other processes use flour and adhesive or even layers of paper.

*Rapid prototyping processes for metals are not currently commercially available*, although they are under development. This means that if a prototype product is being made to tested out a design, it may not have the same mechanical properties as the material that will be used in the final product.

# Fused Deposition Modelling (FDM)

***Fused Deposition Modelling (FDM)*** is the most common form of rapid prototyping process. This is the process used in 3D printers. It is used to make products from polymers such as ABS, PLA, polyamides and nylon.

The steps involved in the process are:

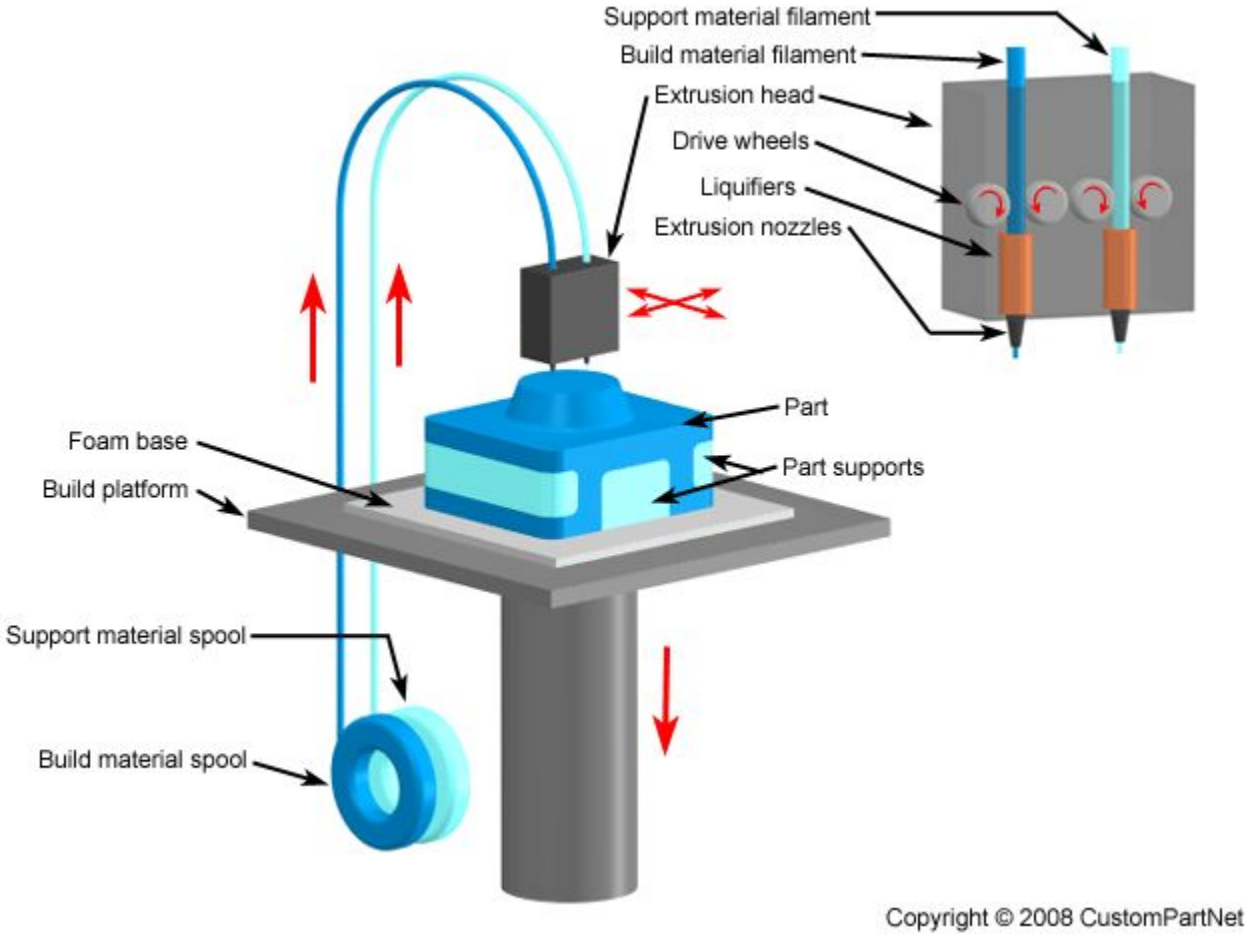
- A 3D CAD model is created of the part, which has been broken down into a series of layers.
- The first layer of polymer is melted and deposited by the printing head, so that it sits on the base. The temperature at which the polymer melts is typically between 200 and 250°C and it quickly cools and solidifies when it is deposited.

# Fused Deposition Modelling (FDM)

- The printing head then rises by the thickness of one layer and deposits the next layer of polymer, building on the first layer.
- This process of depositing a layer and rising continues until the full thickness of the product has been deposited.
- The finished item can then be removed from the base.

# Fused Deposition Modelling (FDM)

***Diagram of the Fused Deposition Modelling (FDM) process:***



# Fused Deposition Modelling (FDM)



# Stereolithography (SLA)

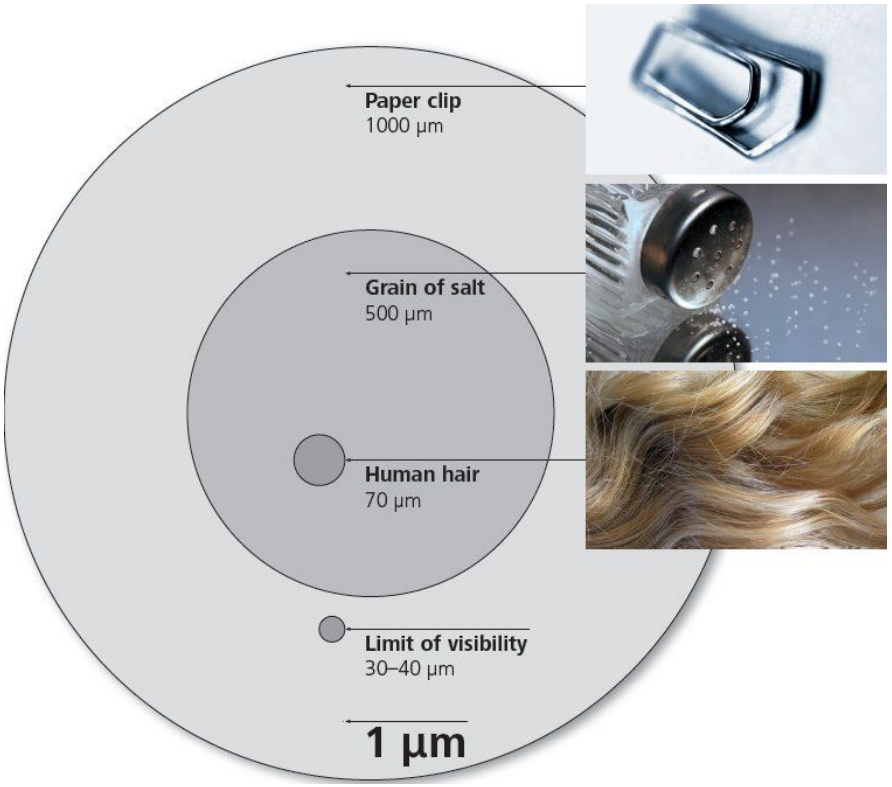
***Stereolithography (SLA)*** is a rapid prototyping process that uses a laser to make polymer products.

The process starts with a 3D CAD model. Computer software is used to break down the drawing into a series of layers (saved in a format called an STL file).

Each layer may be only a few microns in thickness. A movable platform is placed in a tank of ***resin***, which is a liquid form of polymer. The laser then creates the product layer by layer.

# What is meant by 1 micron?

A micron is a metric unit of measurement where one micron is equivalent to **one one-thousandth of a millimetre** [1 micron (1μ) = 1/1000 mm] or 1 micron (micrometer) = 1/1,000,000 of a metre.

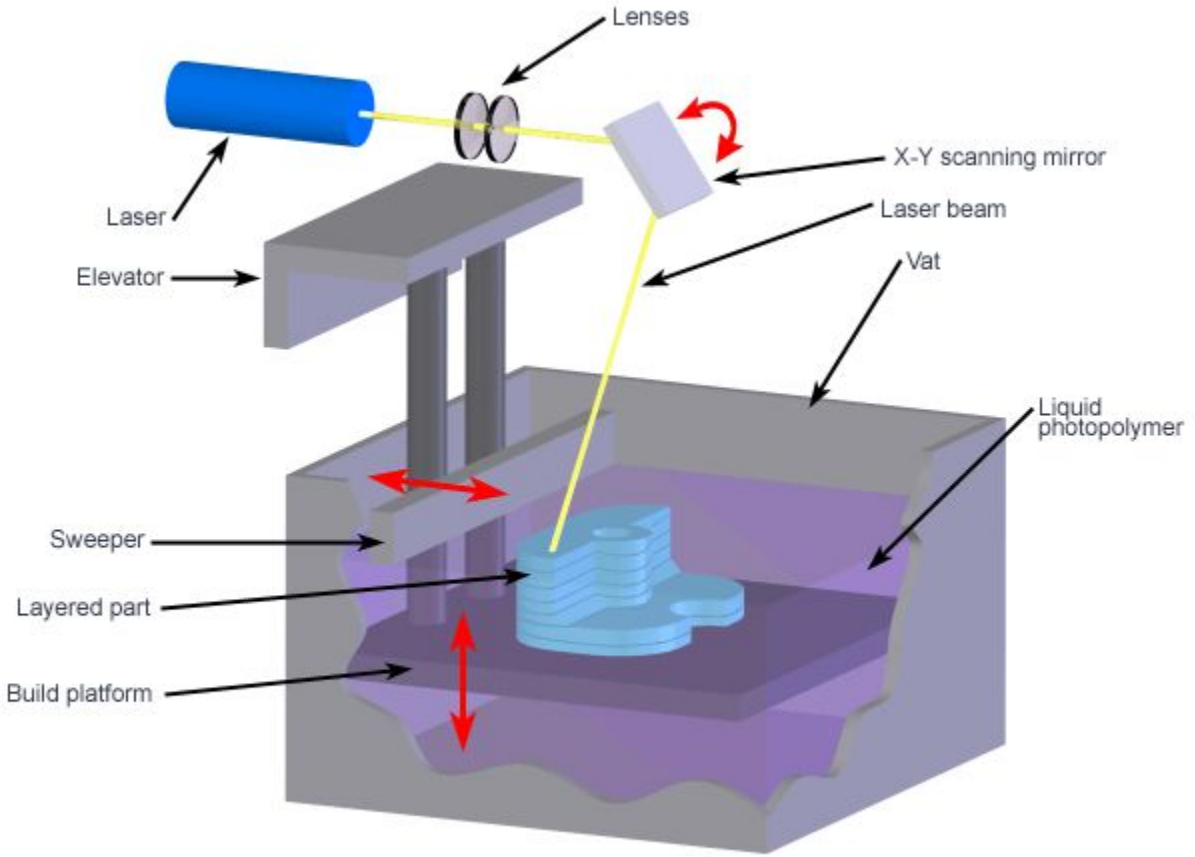


# Stereolithography (SLA)

- The laser draws the shape of the first layer in the resin, so that it sits on the movable plate.
- The laser draws wherever the material is needed, causing the resin to turn into a hard polymer in those places.
- The movable plate is then lowered by the thickness of one layer.
- The laser then draws the second layer in the resin, so that it builds on the first layer.
- The process of drawing and lowering continues until every layer has been drawn, and the full thickness of the product has been created.
- The final part can then be raised above the resin bath and any unhardened resin allowed to drain away.

# Stereolithography (SLA)

***Diagram of the Stereolithography (SLA) process:***




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# Stereolithography (SLA)



# Key Points

- *Additive manufacturing processes make products by adding material where it is needed.*
- *Sintering uses heat and pressure to make solid parts from metal powder.*
- *Rapid prototyping uses additive manufacturing methods to make a product in a single operation.*
- *The starting point for a rapid prototype is a 3D CAD model which is divided into a series of layers by software.*
- *Fused deposition modelling builds up a part using polymer deposited by a printing head.*
- *Stereolithography builds up a part using a polymer from a resin bath that is hardened by a laser.*

# Task: Check your Knowledge and Understanding

- ***Explain*** what is meant by ***additive manufacturing***.
- State ***two*** factors that influence the strength of a metal part made by ***sintering***.
- Name ***two*** materials that can be used to make a part by ***fused deposition modelling***.
- Using a diagram describe the process of making a part using ***stereolithography***.
- ***Explain*** how rapid prototyping may change the way we manufacture products in the future, and ***how*** this may affect our lives.